

CHM301-Quantum Chemistry I

First Mid-semester examination

1st September 2025

Time: 2-3pm

Venue: LH4

Total marks: 15

All the best

1. Calculate the number of standing waves per unit volume in a black body cavity, in the frequency range between 10 Hz and 11 Hz. Verify the statement that "in order to have such a standing wave, the volume of the cavity should be astronomically large". 3 marks
2. Materials can undergo excitations by absorbing a single photon or multiple photons, the latter typically happens when we use intense laser. Based on this information, comment on the statement "photoejection happens only beyond a threshold frequency irrespective of the intensity". 3 marks
3. The series limit of a specific series in the H atomic spectrum is $R_H/9 \text{ cm}^{-1}$. Calculate the wavenumber of the second line in the series. ($R_H = 109680 \text{ cm}^{-1}$) 2 marks
4. Calculate the radius of H atom in the 6th excited state? Do you think that in this state the electron will be bound to the nucleus? Justify your answer. (Bohr radius $a_0 = 0.53 \text{ \AA}$ and the ground state energy is -13.6 eV.) 2 marks
5. Check whether the following function is acceptable in the mentioned interval. If not, try to provide a condition that can make this acceptable (other than changing the interval). 3 marks

$$e^{-x^2} \sum_{n=0}^{\infty} C_n x^n \quad (-\infty, \infty)$$

6. Calculate the expectation value of position for the first excited state of particle in a 1D-box and compare that with the position with highest probability density. What is the value of the maximum probability density? (Answer all the questions mathematically) 2 marks

CHM301-Quantum Chemistry I
Second Mid-semester examination
9th October 2025
Time: 2-3pm
Venue: LH6
Total marks: 35
All the best

1. Check whether $Y_1^1 + Y_1^{-1}$ is an eigen function of $\hat{L}_z$ and $\hat{L}^2$ operators. (3 marks)
2. What will happen if you apply $\hat{L}_+$ operator on Y_1^1 ? First answer the question logically and then verify mathematically. (3 marks)
3. None of the rigid rotor wave functions are eigen functions of $\hat{L}$ operator. Comment on the accuracy of this statement. (3 marks)
4. Assume that there is a potential energy V for rigid rotor. Write down the expression for total energy and ground state wave function. (3 marks)
5. Solve the associated Legendre equation (θ equation after making the substitution $x = \cos\theta$), for $J = 0$ and $m=0$. Hint: Integrate to solve the equation and then check what condition will make the wave function acceptable. (3 marks)
6. Calculate the value of the Y_1^0 wave function of a diatomic molecule with bond length $2\text{\AA}$, at the cartesian coordinates $(0,0,2)$, $(0,0,-2)$, $(0,2,0)$, $(0,-2,0)$, $(2,0,0)$ and $(-2,0,0)$. Draw the cartesian coordinates, mark these points, and write the corresponding wave function values nearby. (3 marks)
7. Solve the harmonic oscillator Schrodinger equation for extremely small displacement and find the form of wave function. No need to normalize. (3 marks)
8. Use the recursion formula and find out an expression for the harmonic oscillator wave function for $v=4$. No need to normalize. (3 marks)
9. Mention three non-classical things that you noticed in quantum harmonic oscillator? (2 marks)
10. Harmonic oscillator approximation is good only at small displacement. Explain why? (3 marks)
11. Ask a question (3 marks)
12. From the diagram given below, write down the expressions for the allowed energy values for a particle in finite square well potential. What can be done to increase the number of bound states? (3 marks)

CHM301-Quantum Chemistry 1
Second Mid-semester examination

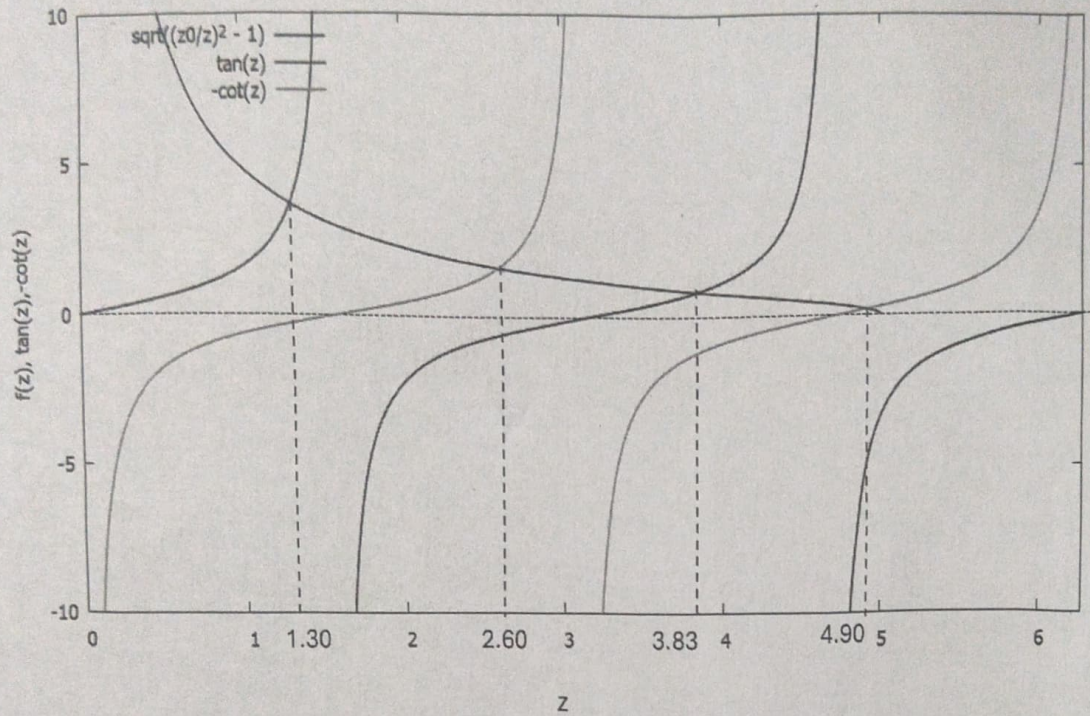
9th October 2025

Time: 2-3pm

Venue: LH6

Total marks: 35

All the best



CHM301-Quantum Chemistry I

End-semester examination

18th November 2025

Time: 2-5 pm

Venue: LH1

Total marks: 50

All the best

- In the H atom Schrodinger equation, we write $\psi = R(r)Y(\theta, \phi)$, and then we replace $\hat{L}^2 Y$ by $\hbar^2 l(l+1)Y$, a result obtained from rigid rotor. Why is it legitimate to assume that the angular functions $Y(\theta, \phi)$ of the hydrogen atom are the same as those for the rigid rotor? Justify this mathematically. Hint: derive the angular equation for H atom. (10 marks)
- Check whether $\psi_{1s} = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$ is an eigen function of the Hamiltonian operator of H atom. (do it mathematically). Find out the eigen value. (3 marks)
- Consider the hydrogen atom and the hydrogen-like ions He^+ , Li^{2+} , and Be^{3+} in their ground states. In which system does the electron have the highest root-mean-square (rms) velocity? Justify mathematically. You may use $\langle T \rangle = \frac{1}{2} m \langle v^2 \rangle$. (5 marks).
- For the H atom ground state, get the expression for the probability $P(\theta) d\theta$ to find the electron between θ and $\theta + d\theta$ for all values of r and ϕ . Then plot $P(\theta)$ vs θ . Remember that the radial probability density function $R^2 r^2$ indicates the probability density on a spherical shell. What does the $P(\theta)$ indicate? Explain with a diagram. (10 marks)
- Calculate the radius of maximum probability of H, He^+ and Li^{2+} and comment on the result. Given:

$$\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{z}{a_0} \right)^{3/2} e^{-zr/a_0}$$
(3 marks)
- Ask one question based on the CHM301 syllabus.
Your question must be conceptual (not a problem-solving or numerical question).
It should be an original question that you genuinely find puzzling or thought-provoking.
Do *not* copy a question from the internet or from any source—marks will be deducted for doing so.
Your question should also be different from the questions already asked in this exam paper. (3 marks)
- For the ground state (1s) of the hydrogen atom (H) and the hydrogen-like ion He^+ , calculate the uncertainty in the radial position (Δr) of the electron. What do you infer? Given:

$$\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{z}{a_0} \right)^{3/2} e^{-zr/a_0}$$
(5 marks).
- Use the Rodrigues formula to derive the radial function of the excited ^{2p}3d state of H atom. (3 marks)
- Calculate the probability to find the 1s electron of H atom beyond 3 Å from the nucleus. (3 marks)
- True/False. Give a one-line justification for each. (5 marks)
 - Orbitals are eigen functions of the angular momentum operator.
 - Orbitals must be real functions.
 - The boundary of H atom is at the Bohr radius.